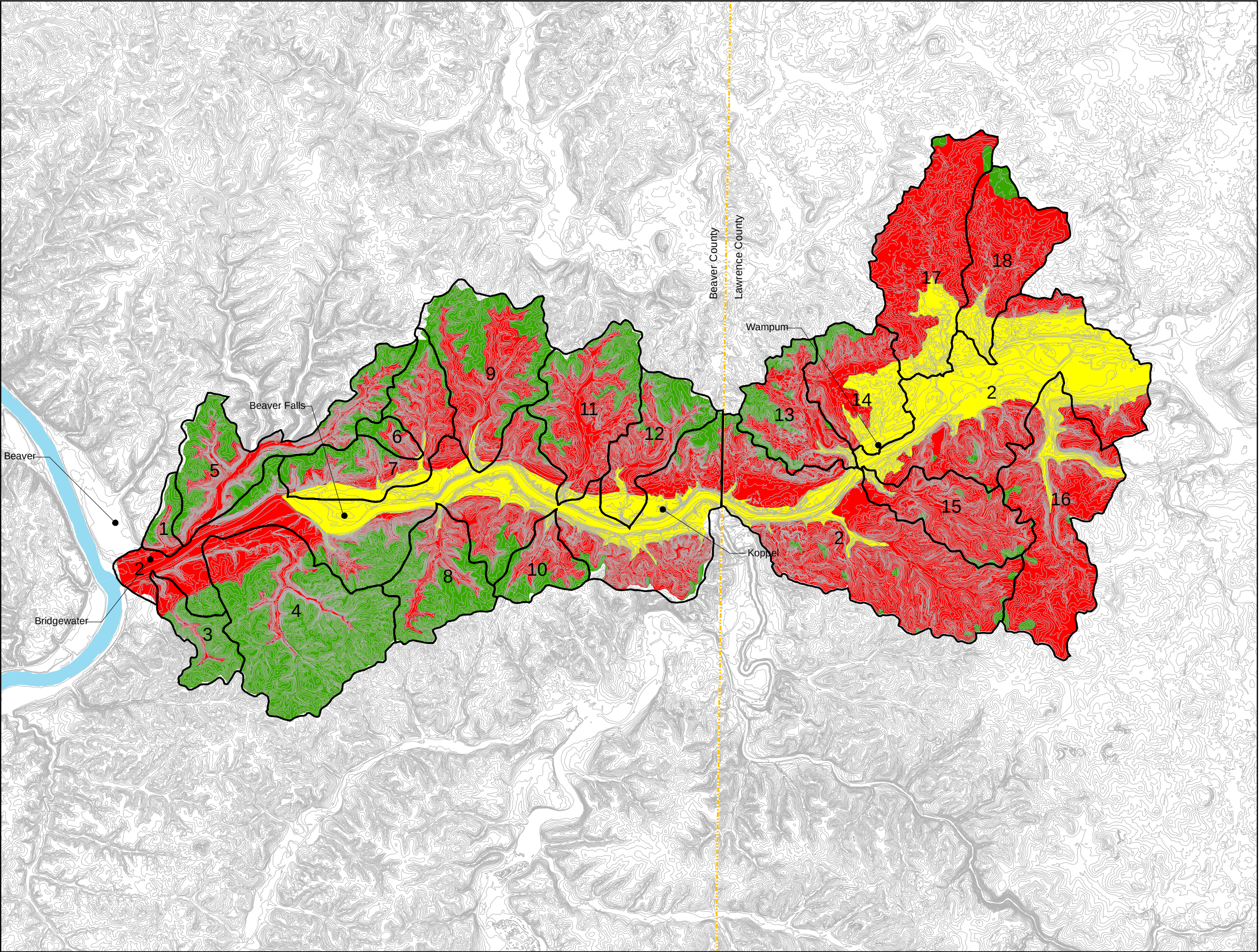




**Legend**

Watershed Boundary

County Boundary

Rivers/Streams/Lakes

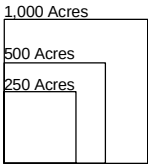
**Primary - Secondary Lithology**

Sandstone - Conglomerate

Sandstone - Shale

Shale - Sandstone

- Subwatersheds**
1. Hamilton Run
  2. Beaver River
  3. McKinley Run
  4. Block House Run
  5. Brady Run
  6. Grimms Run
  7. Walnut Bottom Run
  8. Bennett Run
  9. Wallace Run
  10. Thompson Run
  11. Clarks Run
  12. Stockman Run
  13. Wampum Run
  14. Eckles Run
  15. Snake Run
  16. McKee Run
  17. Jenkins Run
  18. Edwards Run



**GEOLOGY**

**BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council

Prepared by: Environmental Planning and Design, LLC

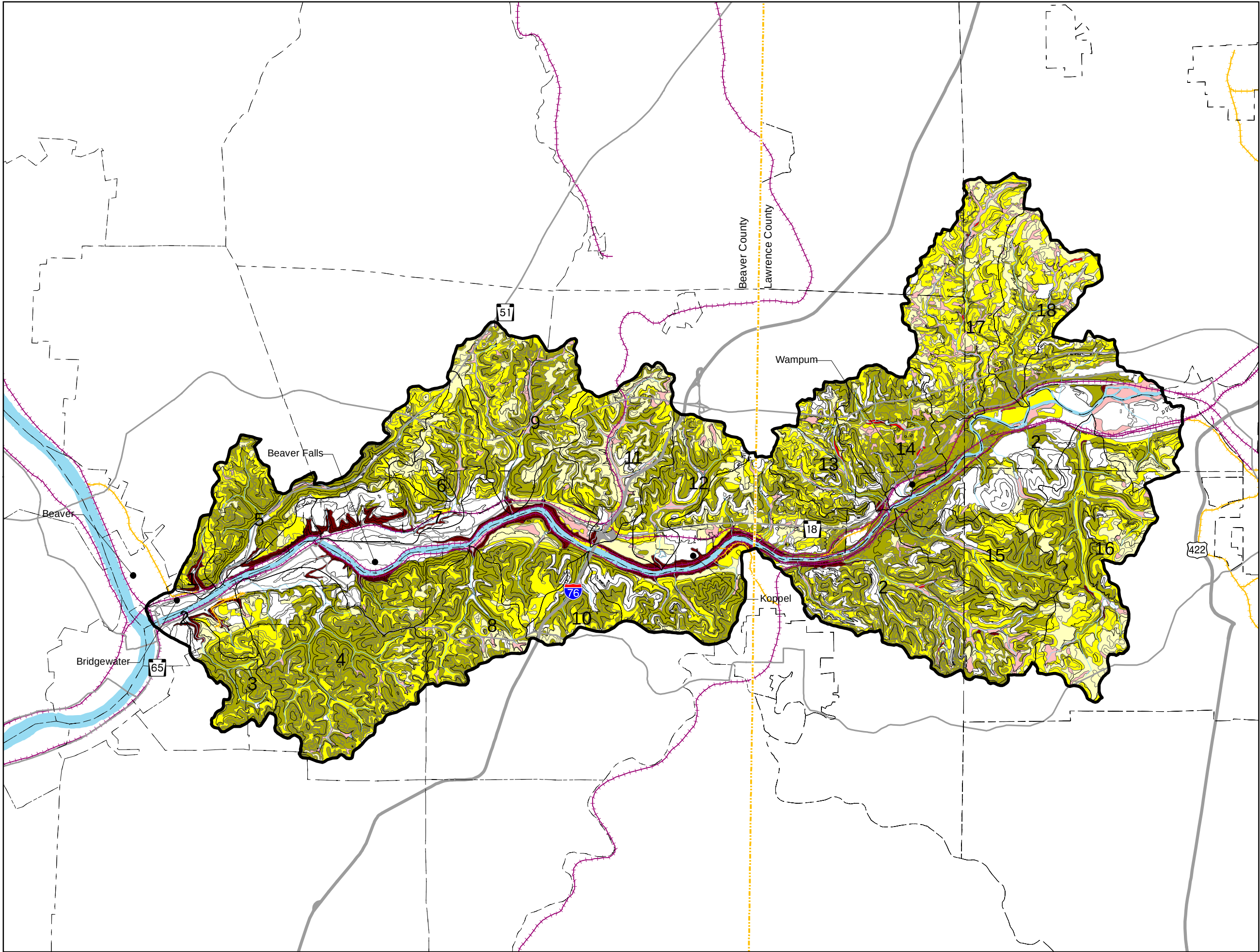


### *Soils*

Soils reveal important characteristics of the land with regard to soil composition, dampness, and susceptibility to erosion. Frequently, soils play an important role in determining a site's ability to support development. Soils vary widely in texture and hue, and more than 10 unique soil types exist in the Beaver River Study Area. These are shown on the Soils-Drainage Map.

It is generally regarded that a number of these soil types are suitable for construction, while others are considered unsuitable for development entirely. A majority of the soils in the Study Area are either moderately to well drained. However, most of the Study Area is characterized by a water table that is between 0 and 15 feet deep, signifying that bedrock and/or the presence of water is generally close to the surface and making the feasibility of development in certain areas potentially more challenging.

More detailed information about the soils can be found in each county's soil survey, produced by the U.S. Department of Agriculture Soil Conservation Service.

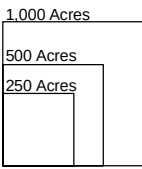


- Legend**
- Study Area Boundary
  - Watershed Boundary
  - County Boundary
  - Municipal Boundaries
  - Rivers/Streams/Lakes
  - Highways
  - Major Roads
  - Active Railroads
  - Inactive Railroads
  - 100' Topographic Contour

- SOIL DRAINAGE CLASSIFICATION**
- Information not available
  - Very poorly drained
  - Poorly drained
  - Somewhat poorly drained
  - Moderately well drained
  - Well drained
  - Somewhat excessively drained

Note:  
Soil drainage information is based on the  
Soil Survey Geographic Data Base,  
United States Department of Agriculture, 2005.

- Sub-Watersheds**
1. Hamilton Run
  2. Beaver River
  3. McKinley Run
  4. Block House Run
  5. Brady Run
  6. Grimms Run
  7. Walnut Bottom Run
  8. Bennett Run
  9. Wallace Run
  10. Thompson Run
  11. Clarks Run
  12. Stockman Run
  13. Wampum Run
  14. Eckles Run
  15. Snake Run
  16. McKee Run
  17. Jenkins Run
  18. Edwards Run



**SOILS - DRAINAGE**  
**BEAVER RIVER CONSERVATION**  
**AND MANAGEMENT PLAN**

Prepared for: Pennsylvania Environmental Council  
Prepared by: Environmental Planning and Design, LLC

Date: August 2006  
1989.06.09



## Land Cover

The Land Cover Map shows the many land uses that exist within the Study Area. Based on a variety of remote sensing techniques, land coverage is depicted using a colored map indicating urban areas of varying densities, open water, vegetative types, and more. High density urban areas are concentrated along the Beaver River within a number of municipalities, including the City of Beaver Falls, Rochester Borough, Rochester Township, New Brighton Township, Pulaski Borough and Wampum Borough, and occupy less than one percent of the Study Area. Daugherty Township, Big Beaver Township, Shenango Township, and North Beaver Township – all in the northern part of the Study Area – have significantly more land devoted to agricultural uses. Finally, despite relatively significant areas of coal mining activity in Shenango Township, Taylor Township, Koppel Borough, West Mayfield Borough, and the City of Beaver Falls, coal mining areas as a whole represent less than four percent of the Study Area. The characteristics of the watershed's land coverage patterns are outlined in the following table:

**Table 4 – 2 : Land Coverage in Study Area**

| Land Cover                               | Acres  | % of Total Area |
|------------------------------------------|--------|-----------------|
| Water                                    | 892    | <2              |
| Emergent and Woody Wetlands              | 45     | <0.1            |
| Coal Mines and Quarries                  | 2,175  | 3.7             |
| Row Crops and Hay Pastures               | 16,428 | 28              |
| Deciduous, Coniferous, and Mixed Forests | 25,310 | 43              |
| Low Density Urban                        | 8,164  | 15              |
| High Density Urban                       | 2,760  | 4.7             |
| Transitional                             | 2,535  | 4.3             |

### Land Coverage Category Descriptions:

Deciduous, Coniferous, and Mixed Forests account for 43 percent of the Study Area, the most acreage of all land coverage categories. This category includes forests found in large, continuous swaths, but also includes smaller stands punctuating or bordering open, agricultural areas.

Row Crops and Hay Pastures represent the majority of agricultural land, amounting to approximately 28 percent of the Study Area. While agricultural lands appear to be relatively well-dispersed, it is worth noting that both the Jenkins Run Watershed and the Edwards Run Watershed share a significant portion of the overall total.

Low Density Urban Areas account for the next greatest amount of coverage, comprising roughly 15 percent of the Study Area. Low Density Urban Areas are difficult to define, but this Plan assumed it to be suburban area development.

High Density Urban Areas account for 4.7 percent of the Study Area and are defined as having intense areas of residential and non-residential development. The areas of highest densities are generally located near old town and borough centers, where buildings are often built close together, or intensely built areas that have been

significantly improved in recent decades to provide automobile access (shopping centers, entertainment facilities, etc).

Transitional Areas are difficult to define using basic remote sensing techniques. This category may include old fields in the process of being reclaimed by woodlands, or, conversely, an old forest in the process of being timbered. Approximately 4.3 percent of the Study Area is characterized as Transitional.

Coal Mines and Quarries only comprise 3.7 percent of the Study Area despite the presence of a few significant mines and quarries highlighted on the map.

Water, and Emergent and Woody Wetlands combined, account for the remaining 2 percent of the Study Area.